

Unit 1 :- Introduction to Microcontroller.

Explain microprocessor and microcontroller in brief.
Compare microprocessor and microcontroller.
Show and explain in brief block diagram of 8051 microcontroller.
List and explain elements of 8051 microcontrollers.
Show pin diagram of 8051 microcontroller
Explain the function of following pins of 8051 microcontroller. (i)PSEN (ii)EA (iii)ALE (iv)RESET
Explain the function of each pin of port 3 of 8051 microcontroller
List and explain applications of 8051 microcontroller.
List and explain features of 8051 microcontroller.
Show and explain register bank of 8051 microcontroller.
Explain register banks of 8051 with its RAM addresses.
Show and explain in brief internal RAM organization of 8051 microcontroller.
Show and explain in brief PSW of 8051 microcontroller.
List and explain any 4 SFRs of 8051 microcontroller with its RAM addresses
Show and explain any Timer SFR of 8051 microcontrollers with its RAM addresses
Show and explain any interrupt SFR of 8051 microcontroller with its RAM addresses
Explain following registers with its RAM address: 1. ACC 2. B 3. DPH 4. SP
Show and explain SCON SFR of 8051 microcontroller with its RAM addresses
Show and explain TCON SFR of 8051 microcontroller with its RAM addresses
Show and explain IE SFR of 8051 microcontroller with its RAM addresses.
Show and explain IP SFR of 8051 microcontroller with its RAM addresses
Explain power down and idle mode of 8051.

Unit 2 :- 8051 Assembly language Programming and I/O port Programmimg.

Identify and correct error in Given assembly code
Explain following addressing modes of 8051 microcontrollers with example any 4.
Identify the addressing modes of any given instructions
Describe f assembler directives with example:
Explain elements of assembly language programming of 8051
Explain Jump and Call instructions of 8051 microcontrollers with example.
Explain Loop instructions of 8051 microcontrollers with example.
Explain the following jump instructions of 8051 microcontrollers with example: (i) SJMP (ii) LJMP
Explain the following Call instructions of 8051 microcontrollers with example: (i) LCALL (ii) ACALL
b) Explain the following 8051 microcontroller instructions: (i) XCH A, @Ri (ii) CJNE A, direct, rel (i)MOVX A,@DPTR (ii)SWAP A
Make use of assembly language instructions to write a program for finding AND, EXOR.
Make use of assembly language instructions to write a program for finding OR, NOT.
Make use of assembly language instructions to write a program to multiply two 8 bit numbers and store the result at 75H.
Construct assembly language program to configure Port 0 as input port.
Make use of assembly language instructions to write a program to find average of 15 numbers.
ADD/SUB/MUL/DIV the numbers 80H and 62H (any 2 numbers) , and show how the Carry , Auxiliary Carry and Parity Flags are affected.
Make use of assembly language instructions to write a program in 8051 to multiply 25 by 10 using the technique of repeated addition

Unit 3 :- 8051 Programming in C

Identify the use of any 3 data types in Embedded C with example.
Identify the use of following Data types in Embedded C: a) unsigned char b) signed int c) sbit d) signed char e) unsigned int f) bit
Identify the use of shift operators in Embedded C with example.
Identify use of Embedded C Programming in 8051.
Identify the use of any 3 logical operators in Embedded C.
Write an 8051 C program to convert packed BCD 0x29 to ASCII and display the bytes on P1 and P2.
Make use of 8051 C program to get a byte of data form P0. If it is less than 100 send it to P1 otherwise send it to P2.
Make use of 8051 C program to toggle all the bits of P0 using the invert operator and P2 using EXOR operator continuously with a 250 ms delay.
Make use of 8051 C program to show use of AND and NOT.
Make use of 8051 C program to show use of shift operators.
Make use of 8051 C program to convert ASCII digits of '4' and '7' to packed BCD and display them on P2
Make use of 8051 C program to get bit P1.0 and send it to P2.7 after complementing it.
Make use of 8051 C program to send values of -4 to +4 to port 1 .
Make use of 8051 C program to show use of any 2 bitwise operators. (4 marks)
Make use of 8051 C program to monitor bit P1.3. If it is low, send 55H to P0; otherwise, send AAH to P2
Identify and correct the error in following 8051 C program to toggle all the bits of P0 continuously: <pre> #include <reg8051.h> void main(void) { for (;;) { P1=0x88; P1=0xAA; } } </pre>
Identify and correct the error in following 8051 C program: <pre> #include <reg52.h> void main() { P1 = FFH; while(1); } </pre>

Unit 4 :- 8051 Memory ,Interrupts and timers/Counters .

Show and explain on chip memory organization of 8051.
Show external RAM interface of 8051
Show and explain any one timer register of 8051.
Explain timer modes of 8051.
Show and explain format of TMOD register.
Explain timers and counters of 8051 in brief.
Explain any 2 interrupts of 8051 in detail.
Explain interrupt priority of 8051 with example.
Explain steps for execution of an interrupt.
Show and explain SCON register.
Explain interrupt priority register of 8051.
Explain following interrupts in 8051: INT0 and INT 1 Timer 0 and Timer 1
Design and show a microcontroller system using 8051 to Interface the external RAM of size 8Kx8.
Design and show a microcontroller system using 8051 to Interface the external RAM of size 16Kx 8.
Design and show a microcontroller system using 8051 to Interface the external RAM of size 32k x 8.
Design and show a microcontroller system using 8051 to Interface the external ROM of size 8Kx8.
Design and show a microcontroller system using 8051 to Interface the external ROM of size 16Kx 8.
Design and show a microcontroller system using 8051 to Interface the external ROM of size 32Kx 8

Unit 5 :- Embedded System.

a) Explain embedded systems and its features.
b) Define embedded system and explain its applications
c) Show and explain components in an embedded system.
d) Show and explain block diagram of embedded system
e) List and explain features of an embedded systems.
f) List and explain application domains and its applications in embedded system
g) Explain following processors in embedded system with its features: : a) GPP b) Single-Purpose Embedded Processor c)Microprocessor d) SoC e) DSP bf Microcontroller
h) Summarize characteristics while designing an embedded system.
i) Summarize challenges of an embedded system
j) Summarize characteristics of an embedded system.
k) Explain challenges while designing an embedded system.
l) Explain RTOS in an embedded system with diagram.
m) Show use of RTOS in embedded system with diagram